

Bar chart in seaborn

I am working with seaborn, and trying to make my bar chart look better.

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

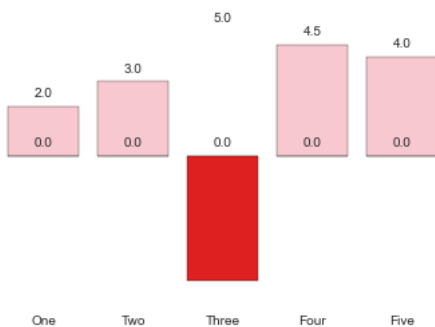
x = ['One', 'Two', 'Three', 'Four', 'Five']
y = [2, 3, 0, 4.5, 4]
y2 = [0, 0, -5, 0, 0]

sns.axes_style('white')
sns.set_style('white')

b = sns.barplot(x,y,color='pink')
sns.barplot(x,y2, color='red')

for p in b.patches:
    b.annotate(
        s='{:.1f}'.format(p.get_height()),
        xy=(p.get_x()+p.get_width()/2.,p.get_height()),
        ha='center',va='center',
        xytext=(0,10),
        textcoords='offset points'
    )

b.set_yticks([])
sns.despine(ax=b, left=True, bottom=True)
```



I actually took code for labelling bars from another thread on stack overflow.

The problem I have that the bar that is negative being labeled on the positive side. I also want to get rid of zeros at the beginning of each graph, and possibly move `x = ['One', 'Two', 'Three', 'Four', 'Five']` in the middle where zeros are, instead of it being on the bottom.

python matplotlib seaborn

edited Sep 11 '15 at 20:11

asked Sep 11 '15 at 16:32

 AK9309
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I think your main problem is that you're calling `barplot` twice. Finding a way to compress it to one call will probably help a lot. Also, provide a link to the question you used to inform this post. – Paul H Sep 11 '15 at 19:12

You should try to limit this to just one question. I'll answer the last one right now: `sns.despine(ax=b, bottom=True, left=True)` will get rid of the frame. – Paul H Sep 11 '15 at 19:17

you're getting the `0.0`'s because you're looping through all of the patches, so you need to include a condition that doesn't annotate when the value of `p.get_height()` isn't greater than zero. – Paul H Sep 11 '15 at 19:21

It's probably much easier to just add the x-axis labels and the annotations as text using `ax.text()` rather than fight with annotations. – iayork Sep 11 '15 at 19:22

you should leave your question as it was -- the solution should stay in the answers sections – Paul H Sep 11 '15 at 20:03

2 Answers

Here's the logic you need make only 1 call to `barplot` and place the annotation on the correct side of the x-axis

```
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```

x = ['One', 'Two', 'Three', 'Four', 'Five']
y = [2, 3, -5, 4.5, 4]

sns.axes_style('white')
sns.set_style('white')

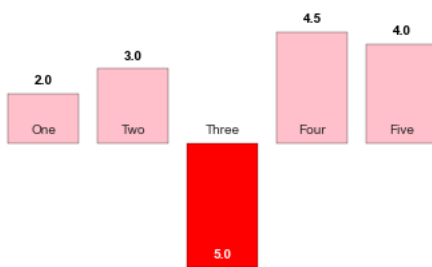
colors = ['pink' if _y >= 0 else 'red' for _y in y]
ax = sns.barplot(x, y, palette=colors)

for n, (label, _y) in enumerate(zip(x, y)):
    ax.annotate(
        s='{:.1f}'.format(abs(_y)),
        xy=(n, _y),
        ha='center', va='center',
        xytext=(0, 10),
        textcoords='offset points',
        color=color,
        weight='bold'
    )

    ax.annotate(
        s=label,
        xy=(n, 0),
        ha='center', va='center',
        xytext=(0, 10),
        textcoords='offset points',
    )

# axes formatting
ax.set_yticks([])
ax.set_xticks([])
sns.despine(ax=ax, bottom=True, left=True)

```



edited Sep 11 '15 at 19:47

answered Sep 11 '15 at 19:27



Paul H
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No need to call `barplot` twice; just pass the list of colors you want as the palette. – [mwaskom](#) Sep 11 '15 at 19:39

1 Also for this kind of plot, much more straightforward to get the bar height out of the data object than out of the axes. – [mwaskom](#) Sep 11 '15 at 19:39

Yeah. Removed the extra call, but opted to set the bar colors after plotting. I suppose generating a palette based on the data would be easy enough. But since we're looping through bars anyway, might not be worth it. – [Paul H](#) Sep 11 '15 at 19:41

@mwaskom eh just went for it – [Paul H](#) Sep 11 '15 at 19:47

Thank you. That's what I was looking for. You can check out my solution. Since I go twice through annotation I can annotations in different positions for negative and positive graphs – [AK9309](#) Sep 11 '15 at 20:04

Thanks everyone for comments. Here is my solution: I have moved the negative number to the end of the list. (Doesn't make that big of a difference though)

```

import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

x = ['One', 'Two', 'Three', 'Four', 'Five']
y = [2, 3, 4, 4.5, 0]
y2 = [0, 0, 0, 0, -5]

figure()
sns.axes_style('white')
b = sns.barplot(x, y2, color='red')

for z in b.patches[4:5]:
    b.annotate(np.round(-z.get_height(), decimals=6), (z.get_x()+z.get_width()/2., -
z.get_height()), ha='center', va='center', xytext=(0, 10), textcoords='offset points',
color='w')

b=sns.barplot(x, y, color='blue')
for p in b.patches[5:9]:
    b.annotate(np.round(p.get_height(), decimals=6),
(p.get_x()+p.get_width()/2., p.get_height()), ha='center', va='center', xytext=
(0, -10), textcoords='offset points', color='w')

```

answered Sep 11 '15 at 20:12



AK9309

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